

Exercise 8: Implementing the Strategy Pattern

CODE:

```
// Strategy Interface

interface PaymentStrategy {

    void pay(double amount);

}

// Concrete Strategy - Credit Card

class CreditCardPayment implements PaymentStrategy {

    public void pay(double amount) {

        System.out.println("Paid Rs." + amount + " using Credit Card.");

    }

}

// Concrete Strategy - PayPal

class PayPalPayment implements PaymentStrategy {

    public void pay(double amount) {

        System.out.println("Paid Rs." + amount + " using PayPal.");

    }

}

// Context Class

class PaymentContext {

    private PaymentStrategy paymentStrategy;

    PaymentContext(PaymentStrategy paymentStrategy) {

        this.paymentStrategy = paymentStrategy;

    }

    public void setPaymentStrategy(PaymentStrategy paymentStrategy) {

        this.paymentStrategy = paymentStrategy;

    }

    public void makePayment(double amount) {

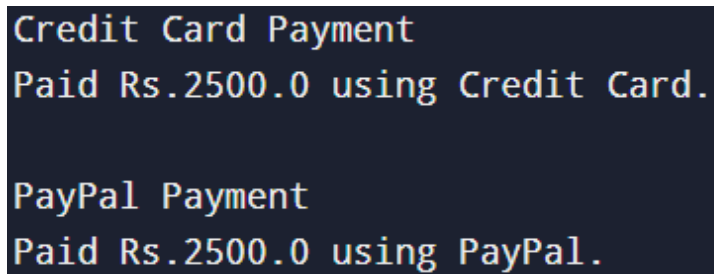
        paymentStrategy.pay(amount);

    }

}
```

```
    }  
}  
  
// Main Class  
  
public class Main {  
    public static void main(String[] args) {  
        PaymentContext payment = new PaymentContext(new CreditCardPayment());  
        System.out.println("Credit Card Payment");  
        payment.makePayment(2500);  
        System.out.println();  
        payment.setPaymentStrategy(new PayPalPayment());  
        System.out.println("PayPal Payment");  
        payment.makePayment(2500);  
    }  
}
```

OUTPUT:

A screenshot of a terminal window with a dark background and light-colored text. The output shows two payment scenarios. The first scenario, 'Credit Card Payment', shows a payment of Rs.2500.0 using a credit card. The second scenario, 'PayPal Payment', shows a payment of Rs.2500.0 using PayPal. The text is formatted with line breaks and indentation to match the code structure.

```
Credit Card Payment  
Paid Rs.2500.0 using Credit Card.  
  
PayPal Payment  
Paid Rs.2500.0 using PayPal.
```